

Giuseppe Baldassarre

Environmental Engineer (Ph.D.) and Forester

Tel. +39 371 361 9920, Email: gi.baldassarre@gmail.com,
[LinkedIn](#), [Google Scholar](#), [ResearchGate](#), [Medium](#), [GitHub](#), [Website](#), [Taleo CV](#)



As a Remote Sensing and GIS Specialist (Ph.D. Environmental Engineering, M.Sc. Mediterranean Forestry), I design and operationalize satellite-based monitoring systems for land, water and agriculture. I combine advanced GIS/RS processing (Python, Google Earth Engine, SEPAL) with decision-support tools and inclusive governance mechanisms to support FAO projects and national partners in climate-vulnerable countries.

Skills

Languages: Italian (native), English, Spanish (fluent), Turkish, Portuguese (intermediate)

Software tools: MATLAB, Python, QGIS, PyQGIS, Office, LATEX, Google Earth Engine, FAO tools (SEPAL, CEO, EarthMap, etc.)

Professional experience

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|---|---|--------------------------|
| Geospatial Analyst | <i>FAO, Land and Water Division</i> | 2023 – present |
| <ul style="list-style-type: none">Leading GEE-to-PRAIS4 automation pipeline for SDG 15.3.1 / Land Degradation Neutrality reporting across 21 UNCCD country Parties (2026 PRAIS4 cycle, COP17 deadline); piloted full workflow for Mongolia (land cover, LPD, SOC, transition matrices)Designed the district-level INFORM Risk Index for FAO Afghanistan's AWARE platform (~401 districts, 60+ indicators); published 35 composite layers to the CKAN catalogueProduced Syria's 10m national land cover map (AlphaEarth Foundations embeddings + Sentinel-2 + Random Forest, GEE) for drought impact assessment (read)Development of the geospatial platform the Gambia GeoLab presented at Geo for Good Summit in Singapore (2025) (FAO news, read)Design and delivery of trainings on "Geospatial Tools for Mangrove Monitoring in The Gambia" (visit) and "Using SEPAL, Google Earth Engine and Collect Earth Online to produce a Land Cover Map of Gambia (2023)" (visit)Flood impact assessment in Pakistan, Oct–Nov 2025 (KP, Punjab, Sindh)Land Cover Atlas of Yemen (read)Geospatial technical assistance for Resilience of Organizations for Transformative Smallholder Agriculture (ROOTS) project, The Gambia (read). Land Cover map of the Gambia (read, watch) | | |
| Post-doctoral researcher | <i>University of Lisbon, Portugal</i> | 2020 – 2023 |
| <ul style="list-style-type: none">Estimating shrub live fuel moisture content for mainland Portugal (read)Study of traditional extensive pastoralism relying on fires in Northern Portugal with GIS/RS analysisMapping Fire Regimes over Portugal (read1, read2) and Turkey (read)Automatic monitoring of the national fuel break network of Portugal with Sentinel-2 in GEE (read) | | |
| EUaid volunteer | <i>Sustainable Rural Development, Vietnam</i> | 2020 – 2022 |
| <ul style="list-style-type: none">Developing an App in GEE to monitor Canopy Cover in Vietnam.Mapping smallholder coffee production system in NW Vietnam. (watch) | | |
| Post-doctoral researcher | <i>Istanbul Technical University, Turkey</i> | 2012 – 2018 |
| <ul style="list-style-type: none">Biomass Burning emission estimation with geostationary observations (read)Description of forest fires over Turkey using the most advanced satellite observations (visit)First version of Forest Fire Monitoring System web-GIS in Turkey (visit) | | |
| Environmental consultant (volunteer) | <i>Aprode Peru (NGO), Peru</i> | 2011 |
| <ul style="list-style-type: none">Field campaigns to outline the solid waste management cycle in rural villages in Northern PeruImplemented a project for the social inclusion of informal recyclers (watch) | | |
| Visiting scholar | <i>University of Wisconsin, USA</i> | 2008, 2009 - 2010 |
| <ul style="list-style-type: none">Development and assessment of advanced satellite fire detection products using statistical toolsTesting product operational applicability over Southern Italy | | |

Education

M.S., Mediterranean Forestry and Natural Resource Management [Erasmus Mundus](#)

1st year: University of Lisbon (Portugal), Valladolid (Spain) & Padova (Italy)

2019

- Natural resource management with decision support systems. Forest governance, agroforestry, fire ecology & management, GIS, research & project development methodology

2nd year: University of Porto (Portugal) & Lleida (Spain)

2020

- Governance of Mediterranean Forests, Economics and Management Non-Profit Organizations, Research Methods, Mathematical Tools For Business and Economics. Environmental Economics.
- Thesis: "Describing fire regimes over Turkey using MODIS fire observations"

Ph.D., Environmental Applications with Remote Sensing

University of Wisconsin, Madison & University of Basilicata

2007 – 2011

- Advanced tools for satellite-based environmental monitoring (biomass combustion emissions)
- Thesis: Performance of satellite techniques for real-time fire monitoring in Italy

B.S. and M.S., Environmental Engineering (summa cum laude)

University of Basilicata (Italy)

2001 – 2004

- Waste-water and waste management; atmospheric physics; water, air and soil pollution dynamics
- Physics, chemistry, geology, statistics, economics

Supporting Documents ([certificates](#), [grades](#), [extra-courses](#), [international cooperation](#), [references](#))